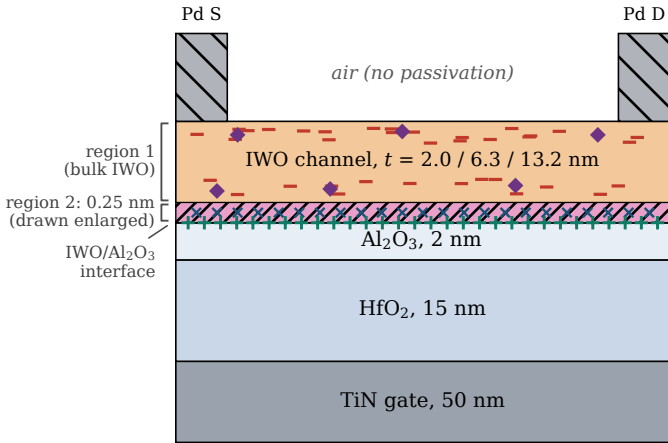




— Acceptor-like band tail, whole IWO film

$$g_{TA} = N_{TA} e^{-(E_c - E)/W_{TA}}, W_{TA} = 40 \text{ meV}$$

$$N_t = N_{TA} W_{TA} = 2.0 / 0.85 / 0.49 \times 10^{19} \text{ cm}^{-3} \text{ (fitted at 2 nm, law in } t)$$

◆ Deep acceptor Gaussian, region 1

$$E_c - 0.6 \text{ eV, width } 0.15 \text{ eV, } 1.3\text{e}+16 \text{ cm}^{-3} \text{ in total (assumed)}$$

✕ Interface acceptor states D_{it} , region 2

$$E_c - 0.30 \text{ eV, peak } 2.9\text{e}+11 \text{ cm}^{-2}\text{eV}^{-1}, 6.4\text{e}+10 \text{ cm}^{-2} \text{ (assumed)}$$

+ Fixed positive charge Q_f at the interface (not a trap)

$$1.73\text{e}+12 \text{ cm}^{-2} \text{ (2.0, 13.2 nm), } 8.7\text{e}+10 \text{ (6.3 nm); fitted on } V_{th,cc}$$

Not modelled: traps in Al_2O_3 and HfO_2 , at the $\text{Al}_2\text{O}_3/\text{HfO}_2$ interface and at the air-side surface; donor-like traps. Background donors $N_D = 2.5\text{e}+17 \text{ cm}^{-3}$ (shallow, not traps). Not to scale.